United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam: : Trimester: Fall - 2024

Course Code: CSE 4889/CSE 489, **Course Title:** Machine Learning, **Sec:** A & B

Total Marks: **40** Duration: **2 hours**

Please answer all the questions. Figures are in the right-hand margin indicates full marks.
Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Question 1: 6+4=10

A. Write and explain the AdaBoost algorithm.

B. How are Instance Weights updated in AdaBoost? Suppose there are 10 instances in a dataset and all the instances have an equal weight one. Among these 10 instances seven are correctly classified and three are misclassified. Now, find the new weights for the correctly classified and misclassified instances. Please show the calculations of every step.

Question 2: 5+5=10

A. Why Ensemble Clustering is needed? Explain different types of Ensemble Clustering?

B. Cluster the following eight points (with (x, y) representing locations) into three clusters:

A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9)

Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2).

The distance function between two points a = (x1, y1) and b = (x2, y2) is defined as-

$P(a, b) = |x2 - x1| + |y2 - y1|$

Use K-Means Algorithm to find the three cluster centers after the first iteration.

Question 3: 5+5=10

A. Describe the approaches to handling overfitting in a deep learning model.

B. Write short notes on the following activation functions:

I. Sigmoid Activation Function
II. Tanh Activation Function
III. ReLU (Rectified Linear Unit) Activation Function
IV. Leaky ReLU Activation Function
V. Softmax Activation Function

Question 4: 5+5=10

A. Describe the role of Convolutional and Pooling layers in CNNs with examples. What are the different types of Pooling? Explain their characteristics.

B. What is Stride? Explain the terms “Zero Padding” and “Valid Padding” in CNNs.